

```

PAITBSCertificate ::= SEQUENCE {
    version                INTEGER ( v3(2) ),
    serialNumber            INTEGER,
    signature               MatterSignatureIdentifier,
    issuer                  Name,
    validity                Validity,
    subject                 MatterPAName,
    subjectPublicKeyInfo    SEQUENCE {
        algorithm           OBJECT IDENTIFIER(id-x962-prime256v1),
        subjectPublicKey     BIT STRING },
    extensions              PAIExtensions }

PAIExtensions ::= SEQUENCE {
    basicConstraint Extension({extnID id-ce-basicConstraints, critical TRUE, extnValue
BasicConstraints {cA TRUE, pathLen 0} }),
    keyUsage Extension({extnID id-ce-keyUsage, critical TRUE, extnValue KeyUsage(<see
text>)}),
    authorityKeyIdentifier Extension({extnID id-ce-authorityKeyIdentifier}),
    subjectKeyIdentifier Extension({extnID id-ce-subjectKeyIdentifier}),
    extendedKeyUsage Extension({extnID id-ce-extKeyUsage}) OPTIONAL
}

```

The PAI certificate SHALL follow the following constraints layered on top of the encoding specified by [RFC 5280](#) within the **TBSCertificate** structure:

1. The **version** field SHALL be set to **2** to indicate v3 certificate.
2. The **signature** field SHALL contain the identifier for signatureAlgorithm **ecdsa-with-SHA256**.
3. The **issuer** field SHALL be a sequence of **RelativeDistinguishedName** s.
4. The **issuer** field SHALL have exactly zero or one **VendorID** value present.
5. The **issuer** field SHALL match, byte-for-byte, the **subject** field of the PAA certificate for the PAA that issued this PAI certificate.
6. The **subject** field SHALL be a sequence of **RelativeDistinguishedName** s.
7. The **subject** field SHALL have exactly one **VendorID** value present.
 - a. If a VendorID value was present in the **issuer** field, the VendorID value found in **subject** field SHALL match the value found in the **issuer** field.
8. The **subject** field SHALL have exactly zero or one **ProductID** value present.
9. The algorithm field in **subjectPublicKeyInfo** field SHALL be the object identifier for **prime256v1**.
10. The certificate SHALL carry the following Extensions:
 - a. Basic Constraint extension SHALL be marked **critical** and have the **ca** field set to TRUE and **pathLen** field set to 0.